HW1:

#1: 3 (b) (d) (e) (h) (i)

4, 7, 8, 10, 11 (a) (c) (d) (e) (g)

#2: 1, 4, 6, 7

# 生成函数.

Generating function

former power series

**Def**: o.p.s.g.f $\{a_n\}_{n\geq 0} \xleftrightarrow{ops} \sum_{n\geq 0} a_n x^n$

**Def** e.p.s.g.f $\{a_n\}_{n\geq 0} \xleftrightarrow{eps} \sum_{n\geq 0} \frac{a_n}{n!} x^n$

**Def**: Dirichlet. $\{a_n\}_{n\geq 1} \xleftrightarrow{Dir} \sum_{n\geq 1} \frac{a_n}{n^s}$

Dirgf

$[x^n] f(x)$. $[\frac{1}{n^s}] f(s)$

Eg: $\{1\}_{n\geq 0} \xleftrightarrow{ops} \frac{1}{1-x}$

$\{1\} \xleftrightarrow{eps} e^x$

$\{1\}_{n\geq 1} \xrightarrow{Dirac} \zeta(s)$

$f(x) \cdot g(x) \triangleq \sum_{n\geq 0} (a_n + b_n) x^n$

$f \circ g = h(x) = \sum_{n\geq 1} f \cdot g = h = \sum_{n\geq 0} \left( \sum_{i=0}^{n} a_i b_{n-j} \right) x^n$

Eg. # of k-tuples $(i_1, i_2, \dots, i_k)$ s.t. $i_1 + \dots + i_k = n$. $(i_j \geq 0)$

fix k $\quad \left\{ \binom{n+k-1}{k-1} \right\} \xleftrightarrow{\text{opsgf}} (1+x+x^2+\dots)^k = \dfrac{1}{(1-x)^k}$

$[x^n]$

$(i_j > 0)$

$\left\{ \binom{n-1}{k-1} \right\} \xleftrightarrow{\text{ops}} \dfrac{x^k}{(1-x)^k} \qquad (x+x^2+\dots)^k = x^k \cdot (1+x+\dots)^k = \dfrac{x^k}{(1-x)^k}$

Cor $k=2$

$\{n+1\}_{n\geq 0} \xleftrightarrow{\text{ops}} \dfrac{1}{(1-x)^2}$

Eg: $\{a_{n+1}\}_{n\geq 0} \xleftrightarrow{\text{ops}} \dfrac{f(x) - a_0}{x}$

$\{a_{n+k}\}_{n\geq 0} \xleftrightarrow{\text{ops}} \dfrac{f(x) - (a_0 + a_1 x + \dots + a_{k-1} x^{k-1})}{x^k}$

$\{n a_n\}_{n\geq 0} \xleftrightarrow{\text{ops}} (xD) f(x)$

$\{n^k a_n\}_{n\geq 0} \xleftrightarrow{\text{ops}} \underline{(xD)^k} f(x)$

$\downarrow$ 미분

$\{P(n) a_n\}_{n\geq 0} \xleftrightarrow{\text{ops}} P(xD) f(x)$ 일 때 식

Cor: $\{n\}_{\geq 0} \xleftrightarrow{\text{ops}} (xD) \dfrac{1}{1-x} = \dfrac{x}{(1-x)^2}$

Eg: $\displaystyle\sum_{n\geq 0} \frac{n^2+2n+9}{n!}$

$\left\{\dfrac{1}{n!}\right\}_{n\geq 0} \xleftrightarrow{\text{ops}} e^x$

$\left\{\dfrac{n^2+2n+9}{n!}\right\}_{n\geq 0} \xleftrightarrow{\text{ops}} \left((xD)^2 + 2(xD) + 9\right) e^x = (x^2+5x+9)e^x$

$\Sigma \dots = f(x=1) = 13e.$

**Thm**: $\{a_n\}_{n\geq 0} \leftrightarrow f(x)$, $\{b_n\}_{n\geq 0} \leftrightarrow g(x)$, $\{c_n\}_{n\geq 0} \leftrightarrow f(x)g(x)$

$$\Rightarrow c_n = \sum_{k=0}^{n} a_k b_{n-k}$$

$1 \leq i \leq m$ $\quad \{a_n^{(i)}\}_{n\geq 0} \leftrightarrow f_i(x)$

$$\Rightarrow \{\sum_{n_1+\cdots+n_m=n} a_{n_1}^{(1)} a_{n_2}^{(2)} \cdots a_{n_m}^{(m)}\} \leftrightarrow \prod_{i=1}^{m} f_i(x)$$

**Cor**: $\{\sum_{i=0}^{n} a_i\} \longleftrightarrow \frac{f(x)}{1-x}$

**Eg**: Harmonic series $\quad H_n = \begin{cases} 0 & n=0 \\ 1+\cdots+\frac{1}{n} & n\geq 1 \end{cases}$

$\{H_n\}_{n\geq 0} \leftrightarrow H(x)$ $\quad$ Find $H(x)$

**Sol**: let $\{a_n\}_{n\geq 0} = 0, 1, \frac{1}{2}, \frac{1}{3}, \ldots$ $\quad \{a_n\} \leftrightarrow A(x)$

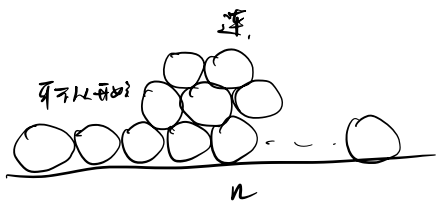
$$\sum_{i=0}^{n} a_i = H_n \Rightarrow H(x) = \frac{A(x)}{1-x}$$

$$A(x) = \sum_{n=1}^{\infty} \frac{x^n}{n}, \quad A'(x) = \sum_{n=1}^{\infty} x^{n-1} = \frac{1}{1-x} \Rightarrow A(x) = -\ln(1-x)$$

$$\Rightarrow H(x) = \frac{1}{1-x} \ln\left(\frac{1}{1-x}\right)$$

A micro-photo of this series

**Eg**: # of "n-fountains" $= f_n$.



Sol:

看递推第2个. $f_n = 1 + \sum_{k=1}^{n-1} (n-k) f_k \longleftrightarrow F(x)$

$$F(x) = \frac{1}{1-x} + C(x)$$

Let $\begin{cases} a_n = n & \longleftrightarrow A(x) \\ b_n = f_n \end{cases}$

$$\Rightarrow C(x) = A(x) F(x) = \frac{x}{(1-x)^2} F(x)$$

$$c_n = \sum_{k=0}^{n} a_{n-k} b_k = \sum_{k=0}^{n} (n-k) f_k$$

$$\Rightarrow F(x) = \frac{1}{1-x} + \frac{x}{(1-x)^2} F(x) \Rightarrow F(x) = \frac{1-x}{1-3x+x^2}$$

Thm: $\{a_n\}_{n\geq 0} \xleftrightarrow{eps} f(x)$, $\{b_n\}_{n\geq 0} \xleftrightarrow{eps} g(x)$

$$\left\{ \sum_{i=1}^{n} \binom{n}{i} a_i b_{n-i} \right\}_{n\geq 0} \longleftrightarrow f(x) g(x)$$

Fact:

① $\{a_{n+1}\}_{n\geq 0} \xleftrightarrow{eps} D(f(x))$

$$f(x) = \sum_{n\geq 0} \frac{a_n x^n}{n!} \qquad Df = \sum_{n\geq 1} \frac{a_n x^{n-1}}{(n-1)!} = \sum_{n\geq 0} \frac{a_{n+1} x^n}{n!}$$

② $\{a_{n+k}\}_{n\geq 0} \xleftrightarrow{eps} D^k(f(x))$

③ $\{n a_n\}_{n\geq 0} \xleftrightarrow{eps} (xD)(f(x))$

④ $\{p(n) a_n\}_{n\geq 0} \xleftrightarrow{eps} p(xD)(f(x))$